

The Physical and Contextual Limits of Quantum Speedup

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Quantum computation is frequently mischaracterized as the simultaneous execution of exponentially many classical computations. This article offers a conceptual clarification of why this “branch-wise parallelism” picture is misleading, demonstrating that the components of a quantum superposition cannot be treated as independently readable classical branches. Quantum speedups arise instead from reversible embeddings of algebraic structure made accessible through engineered interference patterns; more precisely, many speedups identify a class in a partition of possible instances rather than reconstructing the full instance. We review this mechanism through several constraints: unitary garbage erasure is impossible, copying and deletion are context-dependent, and contextuality obstructs a single global classical history. We also distinguish circuit or unitary universality from Turing universality: dense generation of unitaries is not the same as symbolic computation over unbounded inputs with recursion, uniformity, and self-reference. In closed unitary dynamics there is no nontrivial absorbing halting state of the classical many-to-one kind; operational termination requires clocks, flags, measurements, open-system records, or external control. Exponential Hilbert-space dimension supplies a geometry for interference and high-dimensional embedding, not unlimited classical readout.

I. INTRODUCTION

A common description of quantum computation says that a quantum computer evaluates a function on many inputs “at once.” The image is seductive: a register is placed in a superposition, a function is applied to all components, and a final measurement extracts the desired answer. This picture is often invoked when discussing Deutsch–Jozsa [1], Simon’s algorithm [2], Shor’s algorithm [3], Grover search [4], and related oracle or Fourier-sampling procedures [5, 6].

Taken literally, however, this picture is wrong. A quantum computer does not return the list of all computed values. The components of a superposition cannot be individually inspected, copied, deleted, or archived as classical alternatives. They are amplitudes in a Hilbert-space vector, and only suitably arranged interference effects become accessible in a final measurement context, where a context is understood as a compatible set of observables, or equivalently a choice of measurement basis in the non-degenerate finite-dimensional case.

The thesis of this article is the following: Quantum speedups are not many classical computations performed in parallel. They arise when the relevant predicate of a task, often a cell in a partition of possible problem instances, can be embedded into a unitary evolution so that it becomes accessible by interference in a chosen measurement context. At the same time, copying and erasure are physically constrained by no-cloning, no-deletion, and Landauer irreversibility, while contextuality prevents the intermediate process from being interpreted as a single global classical history or as a noncontextual value assignment.

A related ambiguity concerns the word “universal.” Circuit or unitary universality means that a finite gate set can approximate a large class of unitary transformations. Turing universality, by contrast, concerns the symbolic simulation of arbitrary effective procedures on inputs of unbounded length. These are not the same notion. A family of quantum circuits may define an algorithm in the complexity-theoretic sense only when a uniform classical or quantum rule generates the appropriate circuit for each input size. A single finite circuit, or even a universal finite gate set, is not by itself equivalent to the class of partial recursive functions.

This formulation is deliberately restrictive. It does not deny quantum speedup. It denies that branchwise classical computation explains it, and it denies that circuit universality should be conflated with Turing universality.

The claim is not meant to settle interpretive questions about Everettian or many-worlds accounts of quantum computation. Advocates of such accounts have argued that quantum algorithms can be understood in terms of computations distributed across branches or worlds [7–9]. The narrower point made here is operational and model-theoretic: the components of a superposition are not independently readable classical records, and the computational advantage is realized only when the global unitary evolution and final measurement context convert phase relations into an accessible statistic.

The paper is therefore mainly clarificatory. Its technical contribution is the leakage construction in Appendix A, which gives a simple unitary interpolation between the Bernstein–Vazirani interference pattern and ordinary value-query behavior by exporting a tunable amount of which-path information to an inaccessible witness.

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II. SUPERPOSITION IS NOT BRANCHWISE READOUT

Let an n -qubit state be written in the computational basis as

$$|\psi\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle.$$

The dimension of the Hilbert space is 2^n , and this exponential dimension is often used to motivate quantum parallelism. But the amplitudes α_x are not classical memory cells. A measurement yields one outcome per run, with probabilities determined by the Born rule. Recovering the amplitudes requires many similarly prepared systems and many measurement settings, that is, tomography.

Thus a superposition can be acted on linearly, but it cannot be read branch by branch. Holevo's theorem gives one precise expression of this limitation. For an ensemble of states on a Hilbert space $\mathcal{H}$, the accessible classical information is bounded above by the Holevo quantity χ , and $\chi \leq \log_2 \dim \mathcal{H}$. Hence, for an n -qubit transmission, the accessible classical information per use is at most n bits [6, 10].

The state vector may contain exponentially many amplitudes, but the accessible classical readout is sharply constrained. This is the first reason why the naive parallelism metaphor is misleading. The superposed components can interfere, but they are not simultaneously available classical facts.

III. THE USEFUL MECHANISM: SPREAD, COMPUTE, DETECT

Many oracle-based and Fourier-sampling quantum algorithms have the schematic form

$$\begin{aligned} |0\rangle^{\otimes n} &\xrightarrow{\text{spread}} \frac{1}{\sqrt{2^n}} \sum_x |x\rangle \\ &\xrightarrow{\text{compute}} \frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} |x\rangle \\ &\xrightarrow{\text{detect}} \text{measurable interference pattern.} \end{aligned}$$

The middle step may be produced by phase kickback. If

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle$$

and the target register $|y\rangle$ is prepared in $|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$, then

$$U_f |x\rangle |-\rangle = (-1)^{f(x)} |x\rangle |-\rangle.$$

The values of $f(x)$ are not read out individually. They are converted into phases. The final measurement is useful only if those phases interfere so that the desired predicate becomes detectable.

This is the essential mechanism: spread, compute, detect. The algorithm spreads amplitude, computes coherently, and then detects a global feature through interference in a chosen measurement context.

Deutsch-Jozsa is the simplest example. After phase kickback, the query register is in the state

$$\frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} |x\rangle.$$

Applying the final Hadamard transform $H^{\otimes n}$ gives the amplitude of $|0^n\rangle$ as

$$\langle 0^n | H^{\otimes n} \left(\frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} |x\rangle \right) = \frac{1}{2^n} \sum_x (-1)^{f(x)}.$$

This amplitude is ± 1 for constant functions and 0 for balanced functions. The promise makes the global predicate visible as complete constructive or destructive interference.

This example also clarifies what the quantum ‘‘acceleration’’ is an acceleration of. It is not, in general, an acceleration of the recovery of a unique classical object. It is an acceleration of a decision problem whose answer is a coarse-grained class. For the original one-bit Deutsch problem, the four functions

$$f : \{0,1\} \rightarrow \{0,1\}$$

are not individually identified. They are partitioned into the two classes

$$\Pi_D = \{\{00,11\}, \{01,10\}\},$$

where the strings denote the ordered value pairs $f(0)f(1)$. Equivalently, the accessible answer is the parity

$$f(0) \oplus f(1),$$

or, in the usual terminology, the distinction between constant and balanced functions. The final interference pattern is therefore not a readout of $f(0)$ and $f(1)$ separately. It is a readout of the partition cell containing f .

This is the natural decision-theoretic form of the mechanism. A problem instance set Ω is supplied with a partition Π , and the task is not to determine $\omega \in \Omega$ but to determine its equivalence class $[\omega]_\Pi$. Quantum speedup is possible only when the oracle and the measurement basis can be arranged so that different cells of Π are represented by distinguishable interference patterns, while distinctions inside a cell are deliberately discarded. Earlier work on quantum state identification used precisely this language for generalized Deutsch-type problems: trivalent functions of a trit were grouped into equipartitions, and the question was whether the corresponding functional classes could be separated by quantum state identifications [11]. In the present terminology, those examples emphasize the same point: the useful answer is a partition label, not a complete table of values.

There is no contradiction between this coarse-graining and the lower-bound results discussed below. Coarse-graining by itself is not enough. The partition must be compatible with an efficiently implementable unitary encoding and with a final measurement context in which the relevant classes are separated by interference. Arbitrary partitions of arbitrary black-box data need not have this property.

Shor's algorithm is a more sophisticated example. Modular exponentiation is computed reversibly, but the algorithm does not read all values of

$$f(x) = a^x \bmod N.$$

Instead, periodicity is sampled through a quantum Fourier transform. The success comes from algebraic structure: translation symmetry becomes spectral concentration.

The same observation explains why generic predicates are not made easy by superposition. For an arbitrary Boolean string

$$b = (b_1, \dots, b_N),$$

the parity

$$b_1 \oplus b_2 \oplus \dots \oplus b_N$$

depends on every input bit. Exact quantum computation of parity requires exactly $\lceil N/2 \rceil$ queries, giving at most a factor-of-two improvement over the deterministic classical query complexity [12]. More generally, polynomial-method bounds show that for total Boolean functions the bounded-error quantum and classical query complexities are polynomially related, ruling out arbitrary exponential speedup for such black-box total functions [13]. Superposition alone is therefore not the resource. Detectable interference structure is.

Appendix A gives a concrete interpolation of this point for the Bernstein–Vazirani problem. A unitary oracle is modified so that a tunable part of the computational-basis path label is exported to an inaccessible witness register. The Boolean function is unchanged and the enlarged dynamics remains unitary, but the accessible interference pattern is progressively destroyed as more path information leaks.

IV. GARBAGE IS NOT ERASED BY UNITARITY

A second misconception concerns garbage. Classical irreversible computation often hides the physical cost of deleting information. Quantum computation cannot do so, because closed-system evolution is unitary:

$$|\psi\rangle \mapsto U|\psi\rangle, \quad U^\dagger U = I.$$

A non-injective classical function cannot be implemented by a unitary acting only on the input register. It must

be embedded reversibly, for example as

$$U_f : |x\rangle |y\rangle \mapsto |x\rangle |y \oplus f(x)\rangle.$$

The input, or an equivalent record, remains present.

If a reversible computation produces

$$|x\rangle |0\rangle |0\rangle \mapsto |x\rangle |f(x)\rangle |g(x)\rangle,$$

the garbage register $|g(x)\rangle$ cannot simply be deleted. It must be uncomputed, exported, measured, or thermodynamically reset.

No isolated unitary erasure. Let $|\phi_1\rangle$ and $|\phi_2\rangle$ be two normalized pure states on a Hilbert space $\mathcal{H}$ which are not identical up to a global phase. Then there is no unitary U on $\mathcal{H}$ and no fixed normalized state $|0\rangle \in \mathcal{H}$ such that

$$U|\phi_1\rangle = U|\phi_2\rangle = |0\rangle.$$

Indeed, unitaries preserve inner products. If both equalities held, then

$$\langle\phi_1|\phi_2\rangle = \langle\phi_1|U^\dagger U|\phi_2\rangle = \langle 0|0\rangle = 1,$$

so $|\phi_1\rangle$ and $|\phi_2\rangle$ would coincide as rays, contradicting the assumption. This is the same elementary isometric obstruction that underlies the no-cloning and no-deletion results: closed reversible dynamics preserves distinguishability and therefore cannot freely merge, duplicate, or discard unknown alternatives.

Consequently, if $|g(x)\rangle_B$ is a garbage register produced by reversible evaluation of a non-constant function g on input x , there is no unitary acting on B alone that resets $|g(x)\rangle_B$ to $|0\rangle_B$ for all x . The garbage can be removed only by (i) uncomputation, using the input register A , which is reversible and redistributes rather than destroys the information; (ii) coupling B to an environment, dissipating the information thermodynamically; or (iii) measurement of B , creating a classical record whose subsequent reset is subject to Landauer's principle.

Bennett's reversible-computation analysis shows how intermediate garbage can be uncomputed [14–16]; it does not make physical erasure free. Landauer's principle locates the cost in reset, coarse graining, or environmental dissipation [17]. Whether the Landauer–Bennett account fully exorcises Maxwell's demon remains a point of philosophical dispute [18–20], but the operational lesson for quantum computation is unambiguous: unitary dynamics does not delete information.

V. COPYING AND DELETION ARE CONTEXTUAL

Classical information can be copied because it is encoded in distinguishable states. Quantum information cannot generally be copied. If a unitary cloner existed with

$$U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle \quad \text{and} \quad U|\phi\rangle|0\rangle = |\phi\rangle|\phi\rangle,$$

then unitarity preserves inner products: on the one side,

$$\langle \psi | \langle 0 | U^\dagger U | \phi \rangle | 0 \rangle = \langle \psi | \phi \rangle \langle 0 | 0 \rangle = \langle \psi | \phi \rangle,$$

while on the other side the cloned outputs give

$$\langle \psi | \psi | \phi \phi \rangle = \langle \psi | \phi \rangle^2.$$

Equating these yields

$$\langle \psi | \phi \rangle (\langle \psi | \phi \rangle - 1) = 0,$$

so $|\psi\rangle$ and $|\phi\rangle$ are either orthogonal or coincide as rays. This is the no-cloning theorem [21, 22]. For mixed states, the no-broadcasting theorem says that a set of density operators can be broadcast if and only if they pairwise commute [23]. The no-deletion theorem gives the dual obstruction: an unknown quantum state cannot be deleted from one copy while leaving another intact [24].

These results identify the boundary of classicality. Within a fixed orthonormal basis one may copy basis labels,

$$|x\rangle |0\rangle \mapsto |x\rangle |x\rangle,$$

but this copies information only relative to that basis; it does not copy an arbitrary unknown state. In algebraic terms, classical copyability is confined to a commuting context.

Here a *context* means a maximal set of mutually commuting observables, equivalently a maximal Boolean subalgebra of the projection lattice. In the nondegenerate finite-dimensional case it corresponds to an orthonormal basis and the projectors generated by it.

Quantum computation therefore manipulates information that is not globally copyable or disposable. Only after a context is selected, typically by measurement or decoherence, do classical records emerge.

VI. CONTEXTUALITY AND THE ABSENCE OF A GLOBAL CLASSICAL HISTORY

Quantum algorithms move between incompatible descriptions: computational basis, phase basis, Fourier basis, stabilizer basis, and measurement basis. These basis changes already express complementarity. A state sharp in one basis is generally delocalized in another.

In Hilbert spaces of dimension at least three, the Kochen–Specker theorem strengthens this point: one cannot assign pre-existing values to all observables independently of the commuting context in which they are measured [25–27]. Operational versions extend contextuality to preparations, transformations, and unsharp measurements [28]. For single qubits the standard Kochen–Specker obstruction for projective measurements is absent; simple hidden-variable representations of projective qubit measurements can be constructed. This qualification is important: generalized measurements can exhibit Kochen–Specker-type contextuality even for a single qubit [29]. Nevertheless, complementarity already

prevents joint sharp values for noncommuting projective observables and so blocks simultaneous classicalization of all measurement bases.

This matters for quantum computation. The intermediate stages of an algorithm cannot generally be interpreted as a single classical trajectory through pre-existing values. The algorithm is a sequence of context changes implemented by unitary transformations and terminated by a final measurement context. What becomes classical is not the whole history, but only the final record. Between preparation and measurement there is no classically charted path, only coherent evolution whose intermediate description is basis-dependent—like Musil’s bridge whose solid piers stand only at the beginning and the end, while the passage through imaginary quantities in between nevertheless leads from real input to real output [30].

Resource-theoretic results support this perspective, but with important qualifications. Stabilizer operations admit efficient classical simulation by the Gottesman–Knill theorem [31, 32]; hence universal quantum advantage in the circuit model requires resources outside the efficiently simulable stabilizer fragment. Depending on the formalism and dimension, these resources may be described operationally as magic states, phase-space theoretically as Wigner-function negativity, or foundationally as contextuality [33–35]. These characterizations are related but not identical. In particular, the clean necessity results connecting contextuality with magic-state quantum computation are most direct in odd-prime-dimensional qudit settings, while the qubit case involves additional subtleties. The stable message is therefore not that contextuality alone is the complete source of speedup. Rather, quantum algorithms exploit transformations between incompatible contexts, and genuine advantage requires resources that prevent the whole computation from being represented as a single efficiently simulable classical stochastic process.

VII. QUANTUM UNIVERSALITY VERSUS TURING UNIVERSALITY

The word “universal” is used in two substantially different senses in computation theory and quantum information theory. Confusing them leads to conceptual errors.

In the quantum-circuit literature, a finite gate set is called universal if circuits over that gate set can approximate arbitrary unitary transformations on n qubits, typically to arbitrary accuracy in operator norm. In this sense, universality is a statement about the dense generation of groups such as $U(2^n)$ or $SU(2^n)$ by finite collections of gates. It is a statement about controllability of finite-dimensional Hilbert spaces.

Turing universality is different. A Turing-universal device is not merely a generator of transformations on a fixed finite-dimensional space. It acts on symbolic inputs of arbitrary length, admits arbitrarily long compu-

tations, supports parsing and composition of programs, and allows self-reference through coding of machines by strings [36, 37]. This is the setting in which partial recursive functions, undecidability, universal simulation, and the halting problem arise.

Thus a finite quantum circuit, by itself, does not define the same kind of object as a Turing machine. A circuit on n qubits implements a unitary transformation

$$U_n : \mathbb{C}^{2^n} \rightarrow \mathbb{C}^{2^n}.$$

It is a finite-dimensional, norm-preserving map. It has fixed width and depth, no unbounded tape, no open-ended input parsing, and no intrinsic recursion. It also has no autonomous mechanism for extending its own computation to arbitrary symbolic length.

A family of circuits $\{C_n\}_{n \geq 1}$ may compute a language or function in the complexity-theoretic sense only when there is a uniform rule generating the circuit C_n from n . That uniformity condition is itself algorithmic and is normally supplied classically, or by a specified higher-level quantum model such as a quantum Turing machine [7, 38, 39]. In this sense, quantum circuit families and quantum Turing machines are not identified merely by the existence of a universal gate set; their equivalence requires explicit uniformity and simulation assumptions.

This distinction is important for the phrase “quantum computable function.” In the finite-circuit model, a circuit computes only a finite input-output relation, usually probabilistically and after measurement. A uniform circuit family computes a sequence of such relations. A quantum Turing machine or another uniform quantum model can be compared to classical Turing computation, but then the usual Church–Turing issue concerns computability, while the quantum advantage concerns complexity. Standard quantum computation is not believed to enlarge the class of computable classical functions beyond the partial recursive ones; rather, it may reduce the resources needed for some computable tasks.

A related point concerns halting. Classical irreversible computation may enter an absorbing terminal configuration: many distinct pre-halting configurations can evolve into one stable halt state. Such many-to-one absorption is not a primitive of closed unitary dynamics. If $U|h\rangle = |h\rangle$ and $U|\psi\rangle = |h\rangle$, then injectivity of U gives $|\psi\rangle = |h\rangle$. Thus a closed unitary may coherently mark, clock, or correlate halting information, but it cannot collapse distinct computational histories into a single absorbing halt state without exporting information to additional degrees of freedom.

This elementary observation is not a no-go theorem against quantum Turing machines. Standard quantum models implement operational termination through halt flags, clocks, external control, or measurement protocols [7, 38, 40, 41]. The point is only that the classical absorbing-state picture is not itself a closed-unitary primitive.

This reinforces the main thesis. Quantum circuit universality is a powerful notion of finite-dimensional con-

trollability and approximation. Turing universality is a notion of symbolic, unbounded, self-referential computation under an explicit uniformity condition. They intersect in uniform models of quantum computation, but they are not identical. Treating “universal quantum computation” as if it automatically meant Turing universality obscures the distinction between unitary dynamics, measurement, and algorithmic control.

VIII. CONCLUSION

Quantum computation is not classical computation performed in exponentially many parallel branches whose results can later be inspected. Its power lies in reversible amplitude manipulation, phase-sensitive computation, and contextual measurement. Speedup appears when the desired predicate—often a partition label rather than a full classical description of the instance—can be encoded into a unitary process and made visible by interference in an appropriate final context.

Exponential Hilbert-space dimension should not be mistaken for exponentially accessible classical storage. Large dimension provides room for structured embeddings and interference, but readout is constrained by measurement, Holevo bounds, and the chosen context. In this sense high-dimensional quantum state space is a resource for controlled recombination, not for unrestricted branchwise inspection.

The negative constraints are just as important as the positive mechanism. Garbage cannot be erased by unitary dynamics; it must be uncomputed or exported, and physical reset is thermodynamic. Unknown quantum states cannot be freely copied, broadcast, or deleted. Incompatible contexts prevent the assignment of a single efficiently simulable classical history to the computation.

Nor should circuit or unitary universality be confused with Turing universality. A finite universal gate set provides dense control over unitary transformations; it is not, by itself, a symbolic machine with arbitrary parsing, recursion, self-reference, and a classical halting semantics. In closed unitary dynamics there is no nontrivial absorbing halt state of the classical many-to-one kind. Termination and output extraction require clocks, flags, measurements, open-system records, or external control.

Thus quantum computation is neither magic parallelism nor ordinary classical computation in disguise. It is context-dependent, thermodynamically embedded, reversible information processing whose useful answers must be engineered as interference phenomena.

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Appendix A: A unitary leakage knob for Bernstein–Vazirani interference

The Bernstein–Vazirani algorithm gives a particularly transparent example of quantum speedup by interference rather than by any literal parallel readout of classical function values [6, 38, 42]. Let $a \in \mathbb{F}_2^n$ be an unknown string and define

$$f_a(x) = a \cdot x \pmod{2}, \quad x \in \mathbb{F}_2^n. \quad (\text{A1})$$

The usual reversible oracle is

$$O_a |x\rangle |y\rangle = |x\rangle |y \oplus f_a(x)\rangle. \quad (\text{A2})$$

With the target register initialized in $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$, this oracle acts by phase kickback,

$$O_a |x\rangle |-\rangle = (-1)^{a \cdot x} |x\rangle |-\rangle.$$

Thus

$$\begin{aligned} |0^n\rangle |-\rangle &\xrightarrow{H^{\otimes n}} 2^{-n/2} \sum_x |x\rangle |-\rangle \\ &\xrightarrow{O_a} 2^{-n/2} \sum_x (-1)^{a \cdot x} |x\rangle |-\rangle \\ &\xrightarrow{H^{\otimes n}} |a\rangle |-\rangle. \end{aligned} \quad (\text{A3})$$

Equivalently, the final amplitude of $|z\rangle$ is

$$2^{-n} \sum_{x \in \mathbb{F}_2^n} (-1)^{(a \oplus z) \cdot x} = \delta_{z,a}. \quad (\text{A4})$$

The hidden string is therefore recovered exactly with one coherent query. The essential step is not the evaluation of many classical values, but the conversion of a global phase character into a computational-basis population by a Fourier transform over $\mathbb{F}_2^n$.

We now introduce a reversible modification that degrades this interference without changing the Boolean function f_a . Let $S \subseteq \{1, \dots, n\}$ have cardinality k , and let $\bar{S}$ denote its complement. For $x \in \mathbb{F}_2^n$, write

$$x_S = (x_i)_{i \in S} \in \mathbb{F}_2^k, \quad x_{\bar{S}} = (x_i)_{i \in \bar{S}} \in \mathbb{F}_2^{n-k},$$

where a fixed ordering of the coordinates in S and $\bar{S}$ is understood. Thus x_S and $x_{\bar{S}}$ denote the subvectors of x supported on S and on its complement, respectively. Define the unitary permutation

$$W_a^S |x\rangle |y\rangle |e\rangle = |x\rangle |y \oplus f_a(x)\rangle |e \oplus x_S\rangle, \quad (\text{A5})$$

where the third register is a witness, garbage, or environmental register. The oracle still computes the same function f_a , and the enlarged map is reversible. However, if the third register is not coherently available to the algorithm, then it carries which-path information about the computational-basis label x . Here “which-path” is meant in the standard interferometric sense: a record relative to a chosen computational basis, not an independently readable classical computation carried by a component of the superposition. In this respect the construction is a tunable version of the familiar which-way suppression of interference [43–46]. It is also closely related to Aaronson’s discussion of garbage in quantum query models, where sufficiently informative garbage in an oracle response is suggested to behave, in appropriate settings, like decoherence or measurement of the response [47].

Claim: partial loss of the Bernstein–Vazirani interference pattern. Let the query register be initialized in $|+\rangle^{\otimes n}$, the answer register in $|-\rangle$, and the witness register in $|0^k\rangle$. Apply W_a^S , trace out the witness register, and then apply $H^{\otimes n}$ to the query register. The final measurement distribution on the query register is

$$\Pr(z) = 2^{-k} \mathbf{1}[z_{\bar{S}} = a_{\bar{S}}]. \quad (\text{A6})$$

Thus one coherent Bernstein–Vazirani-style query determines the $n - k$ components of a outside S with certainty, while the observed coordinates z_S are uniformly distributed over $\mathbb{F}_2^k$ and carry no information about a_S . In particular, the usual one-query speedup survives only on those coordinates whose computational-basis path labels have not been exported to the inaccessible witness.

Operationally, after this first query one may recover the remaining coordinates by k ordinary value queries on the standard basis vectors e_i , $i \in S$, since $f_a(e_i) = a_i$. This gives a natural query count $1 + k$ for $k < n$. At the endpoint $k = n$, the first Bernstein–Vazirani-style query gives the uniform distribution and is best omitted, so the standard n value queries suffice. Hence this simple strategy interpolates as

$$Q_{\text{strat}}(n, k) = \begin{cases} 1 + k, & 0 \leq k < n, \\ n, & k = n. \end{cases} \quad (\text{A7})$$

The point needed here is not a separate optimality theorem for arbitrary adaptive use of the leaked channel, but the underlying structural fact proved below: the phase-interference channel available after tracing out the witness contains no dependence on a_S . Therefore no strategy using only this dephased interference channel can extract a_S ; after $a_{\bar{S}}$ has been recovered, at least k independent value queries are needed to determine the remaining k -bit string a_S .

Proof. After the first Hadamard transform and the application of W_a^S , the joint state is

$$|\psi_a^S\rangle = 2^{-n/2} \sum_{x \in \mathbb{F}_2^n} (-1)^{a \cdot x} |x\rangle |-\rangle |x_S\rangle. \quad (\text{A8})$$

Tracing out the witness register gives the reduced density operator on the query register

$$\rho_a^S = 2^{-n} \sum_{\substack{x, x' \in \mathbb{F}_2^n \\ x_S = x'_S}} (-1)^{a \cdot (x \oplus x')} |x\rangle \langle x'|. \quad (\text{A9})$$

Hence coherences remain only between computational-basis components that agree on the coordinates in S . The witness register has therefore implemented a perfect which-path record for those coordinates.

Now apply $H^{\otimes n}$ and measure in the computational basis. For $z \in \mathbb{F}_2^n$,

$$\Pr(z) = \langle z | H^{\otimes n} \rho_a^S H^{\otimes n} | z \rangle \quad (\text{A10})$$

$$= 2^{-2n} \sum_{\substack{x, x' \in \mathbb{F}_2^n \\ x_S = x'_S}} (-1)^{a \cdot (x \oplus x') + z \cdot (x \oplus x')}. \quad (\text{A11})$$

Set $t = x \oplus x'$. The constraint $x_S = x'_S$ is equivalent to $t_S = 0$. For each such t , there are 2^n choices of x . Therefore

$$\Pr(z) = 2^{-n} \sum_{\substack{t \in \mathbb{F}_2^n \\ t_S = 0}} (-1)^{(a \oplus z) \cdot t} \quad (\text{A12})$$

$$= 2^{-n} \sum_{t_S \in \mathbb{F}_2^{n-k}} (-1)^{(a_S \oplus z_S) \cdot t_S}. \quad (\text{A13})$$

By character orthogonality over $\mathbb{F}_2^{n-k}$,

$$\sum_{t_S \in \mathbb{F}_2^{n-k}} (-1)^{(a_S \oplus z_S) \cdot t_S} = \begin{cases} 2^{n-k}, & z_S = a_S, \\ 0, & z_S \neq a_S. \end{cases} \quad (\text{A14})$$

This proves Eq. (A6).

It remains to identify what information remains available through the interference channel. Consider the effective reduced oracle obtained from one call to W_a^S followed by tracing out the witness register. For the phase-kickback use of the answer register, this reduced channel on the query register is

$$\mathcal{E}_a^S(\rho) = D_a \left(\sum_{s \in \mathbb{F}_2^k} P_s \rho P_s \right) D_a^\dagger, \quad (\text{A15})$$

where

$$D_a = \sum_x (-1)^{a \cdot x} |x\rangle \langle x| \quad (\text{A16})$$

and

$$P_s = \sum_{x: x_S = s} |x\rangle \langle x|. \quad (\text{A17})$$

Equivalently, in matrix elements,

$$\mathcal{E}_a^S(\rho) = \sum_{\substack{x, x' \\ x_S = x'_S}} (-1)^{a \cdot (x \oplus x')} \rho_{xx'} |x\rangle \langle x'|. \quad (\text{A18})$$

Inside each block $x_S = x'_S = s$, the contribution of a_S cancels:

$$a \cdot (x \oplus x') = a_S \cdot (x_S \oplus x'_S). \quad (\text{A19})$$

Therefore the reduced phase-interference channel depends on a only through a_S . The hidden components $a_{\bar{S}}$ are not merely hard to extract by the final Hadamard transform; they are absent from the accessible phase channel once the witness has been traced out.

The same dephasing in the x_S basis occurs for any preparation of the answer register, since the witness records x_S independently of the answer-register state. Consequently, once a_S has been obtained by the surviving Bernstein–Vazirani interference, the remaining components $a_{\bar{S}}$ must be learned, if at all, through ordinary value information in

$$f_a(x) = a_S \cdot x_S + a_{\bar{S}} \cdot x_{\bar{S}}. \quad (\text{A20})$$

Since $a_{\bar{S}}$ is then known, a standard value query supplies one linear equation over the unknown vector $a_S \in \mathbb{F}_2^k$. Querying the standard basis vectors e_i , $i \in S$, recovers these k components directly. This proves the stated interpolation strategy in Eq. (A7). $\square$

The two extremes make the physical point explicit. For $k = 0$, no computational-basis path label is exported and the ordinary one-query Bernstein–Vazirani interference pattern is recovered. For $k = n$, the witness records the entire computational-basis label x , all off-diagonal terms in that basis are removed from the algorithm’s accessible state, and the final Hadamard transform has no global phase pattern left to recombine. The Boolean function f_a has still been evaluated coherently by a unitary permutation on the enlarged system. What has been lost is not evaluation but recombination.

This example shows that reversibility alone is not the relevant criterion. The map W_a^S is a unitary permutation on the enlarged Hilbert space. If the witness register remains coherently accessible, one may uncompute it and restore the full interference pattern. If, however, the witness register has escaped into an inaccessible environment, then the reduced dynamics available to the algorithm is no longer reversible. The witness register is precisely the uncomputed garbage identified in the main text as an obstruction to free unitary erasure, here promoted to a tunable knob. The speedup is lost not because the Boolean function f_a has changed, nor because the global dynamics has ceased to be unitary, but because an inaccessible which-path record selects a computational-basis context incompatible with the later Fourier-basis recombination.

Pointedly stated, this demonstrates that a quantum computer does not “compute all branches at once and let us read them”—where the scare-quoted branch language names precisely the classical picture being rejected. If it did, leaking the computational-basis path labels to a witness register would not matter: the function $f_a(x)$ was still computed. Instead, the appendix shows that

the speedup relies on *recombining amplitudes* by interference. When uncomputation is blocked by an inaccessible witness, the accessible dynamics is dephased in the com-

putational basis, the interference is quenched, and the lost coordinates must be recovered, if at all, by ordinary value queries.

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